

Research Papers

Energy management and capacity allocation method of hybrid energy storage system based on port transportation-energy coupling characteristics[☆]

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ABSTRACT

To promote the consumption of renewables in ports, based on the transportation-energy coupling characteristics of ports, a nested bi-layer energy management and capacity allocation method of hybrid energy storage system (HESS) is proposed to coordinate the imbalance between hydrogen/ electricity supply and demand. First, to coordinate the imbalance between multi-energy supply and demand, a multi-energy load model based on traffic scheduling in ports is established to realize the demand-side management of load. Then, considering carbon emission costs of ports and degradation costs of HESS, a “transportation-energy” collaborative scheduling model is proposed with the goal of optimizing the daily operating costs of ports. Furthermore, a HESS capacity allocation model is formulated with the goal of minimizing the annual operational life cycle cost of ports. Finally, to improve the accuracy and efficiency of solving the capacity allocation model, an improved model-pursuing sampling (MPS) algorithm based on the metamodel is presented. The simulation results based on port historical data show that, compared with the traditional scheduling method, the daily operating cost of the collaborative scheduling method is reduced by 7.2 %. Compared with empirical capacity, under the optimal HESS capacity, the annual cost of the port is reduced by 8.2 %. The annualized cost of ports obtained under the improved MPS algorithm is lower.

1. Introduction

At present, seaports undertake most of the logistics transit in the world, while also leading to significant energy consumption and environmental pollution [1]. Port equipment electrification and hydrogenation are an important means for port energy consumption transformation. It can improve the energy efficiency, reduce carbon emissions, and achieve sustainable transportation of ports [2]. In recent years, cold ironing technology has been rapidly developed [3]. At the same time, fixed transportation equipment such as quay cranes (QCs) have gradually realized electrification, and mobile equipment, such as container trucks, yard cranes (YCs), and rubber-tired gantry (RTG) cranes are also gradually becoming hydrogen powered [4]. With the

electrification/hydrogen power of the key logistics equipment in port, the traffic scheduling of ports not only affects logistics operations but also determines changes in port energy consumption, forming the “transportation- energy” coupling characteristics of ports.

Studies have shown that renewable energy will become the most important energy source for low-carbon or even zero carbon ports in the future [5]. In addition, if ports can realize the localized production and consumption of hydrogen energy through renewables, it can effectively utilize the efficient and clean advantages of hydrogen energy and reduce costs, further promoting clean energy consumption in ports. However, with the increase of the penetration rate of renewables in ports, the anti-peak regulation characteristics of renewables seriously influence the “source” and “load” balance of port energy, affecting the clean energy utilization of ports. Energy storage systems (ESSs) are an effective way

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Nomenclature

QC, YC, RTG	Quay crane, yard crane, rubber-tired gantry crane
TEU	Twentyfoot equivalent unit
B-QC	Berth and quay crane
ESS, HESS, P2H	Energy storage system, hybrid energy storage system, power to hydrogen
LiB, Elz	Lithium-ion battery, electrolyzer
SOC, LOH, DOD	State of charge, level of hydrogen, depth of discharge
CS, SS	Conventional scheduling, simultaneous scheduling
MPS, PSO	model-pursuing sampling, particle swarm optimization
v, q, y	Set of vessels, quay cranes, container yards
v_{cra}, p_{cra}	Loading efficiency (TEU/h), power of a quay crane (kW)
$n_{QC,i}$	Number of quay cranes allocated to vessel i
$P_{iron,i}, P_{cra,i}$	Cold ironing power (kW) and quay cranes power (kW) of vessel i
$l_{safe}, l_{line}, l_{ves,i}$	Safety distance (m) between vessels, length (m) of berth coastline, length (m) of vessel i
$x_{sta,i}, x_{end,i}, l_{ves,i}$	Start (m), end (m) of vessel i berth location
$t_{st,i}, t_{et,i}, t_{wait,i}$	Start time (h), end time (h) and waiting time (h) of berth of vessel i
$t_{ber,i}$	Time (h) of vessel i in berthing
$T_{esti,i}, T_{wait}^{min}, T_{wait}^{max}$	Expected vessel berth start time (h), minimum and maximum waiting time (h) for a vessel berth
$n_{QC,i}^{min}, n_{QC,i}^{max}$	Maximum and minimum quay cranes assigned to vessel i
$n_{QC}^{max}, x_{QC,m}$	Total number of quay cranes, No. m of quay cranes
n_{TEU}^{max}	Maximum number of containers that can be stored in a single yard
A_j	Yard location
D_j	Transportation distance (m)
$n_{TEU,j}$	Number of containers of vessel i at the corresponding yard location A_j
$S_{m,i}(k)$	State variable. 1 means that vessel i is served by the quay crane m at time k , otherwise is 0
$S_{i,j}$	State variable. 1 means that the j -th yard serves vessel i .
v_{trk}, v_{RTG}	Hydrogen consumption per unit distance of container trucks (Nm^3/km) and per unit container (Nm^3/TEU)
$V_{trk,i}$	Hydrogen consumption per unit time (Nm^3/h) of container trucks that serving vessel i
$V_{RTG,i}$	Hydrogen consumption of RTG that serving vessel i per unit time (Nm^3/h)
T_d	Control cycle of day-ahead scheduling (h)
C_{ope}	Daily operating costs in port areas (\$)
$C_{bat}, C_{elz}, C_{ves}, C_{grid}$	Cost of LiB (\$), Elz (\$), vessels (\$), main grid (\$)

$C_{cap,x}(x=bat, elz)$	Capital cost of LiB, Elz (\$)
$C_{om}, C_{stup}, C_{shud}$	Maintenance cost (\$), start-up cost (\$), shutdown cost (\$)
$\delta_i, \gamma_{on}, \gamma_{off}$	On/off state, start up and shutdown state of Elz
P_{bat}, P_{elz}	Power of LiB (discharge is positive, in kW) and Elz (the value is negative, in kW)
E_{bat}, W_{H^2}	Energy capacity (kWh) of LiB, capacity of hydrogen tank (Nm^3)
η_{LiB}, η_{elz}	Energy conversion efficiency of LiB, Elz
$C_{penal}, C_{penal}^{ves}$	Penalty cost per unit capacity of the main grid (\$/kWh) and vessels (\$/h)
$c_{co2, berth,i}, c_{co2, grid}$	CO ₂ emission cost per unit capacity of vessel i waiting for berth (\$/kg) and the main grid (\$/kg), respectively
C_{grid}, C_{co2}	Unit cost of the main grid (\$/kWh), carbon emission cost per capacity (\$/kg)
α_{co2}	CO ₂ emission factor of the main grid (kg/kWh)
$P_{bat}^{min}, P_{bat}^{max}, P_{elz}^{min}, P_{elz}^{max}, P_{grid}^{min}, P_{grid}^{max}$	Minimum and maximum power of LiB (kW), Elz (kW) and the main grid (kW)
$SOC_{bat}^{min}, SOC_{bat}^{max}, LOH_{tank}^{min}, LOH_{tank}^{max}$	Minimum and maximum capacity of LiB and hydrogen tank
$\kappa_{st}^{bat}, \kappa_{et}^{bat}, \kappa_{st}^{tank}, \kappa_{et}^{tank}$	Constraint coefficient of capacity at the beginning and end of day
$P_{port}, P_{grid}, P_{pv}, P_{wt}$	Power of port (kW), grid (the power input to the port is positive, in kW), PV (kW), wind turbine (kW)
$T_{lft, LiB}, T_{lft, elz}$	Total service time of LiB (a) and Elz (a)
$T_{day, LiB}, T_{day, elz}$	Daily degradation of LiB and Elz
$T_{lt, LiB}, T_{lt, elz}$	Lifetime of LiB and Elz
$C_{port}, C_{hess}, C_{run}$	Annual cost of ports (\$), HESS (\$) and the operating cost of ports (\$)
C_{pb}, C_{eb}	Unit power capacity cost (\$/kW) and unit energy capacity cost (\$/kWh) of LiB
C_{pe}, C_{ee}	Unit power capacity cost (\$/kW) of Elz, unit energy capacity cost of hydrogen tank (\$/Nm ³)
γ, T_{rt}	Interest rate, project operation period (a)
n_{bat}, n_{elz}	The replacement times of LiB and Elz in T_{rt}
$C_{cap}, C_{repl}, C_{maint}, C_{rec}$	Capital cost (\$), replacement cost (\$), maintenance cost (\$), and remanent value (\$) of HESS
$C_{pb,r}, C_{eb,r}, C_{pe,r}$	Remanent values of unit power (\$/kW) and unit energy (\$/kWh) of LiB, remanent value (\$/kW) of unit power of Elz
$C_{bat}^{maint}, C_{elz}^{maint}, C_{tank}^{maint}$	The annual maintenance costs of unit capacity of LiB (\$/kWh), Elz (\$/kWh) and hydrogen tank (\$/Nm ³)
φ	Annual service coefficient of HESS

to coordinate the imbalance between renewable energy and load [6]. However, with the acceleration of the integration of port transportation and energy, port energy consumption is deeply influenced by logistics characteristics, which leads to greater challenges to the coordinated control of ESSs.

The key to coordinating the supply and demand imbalance of hydrogen/electricity in ports through ESSs mainly includes two aspects: one is the coordination strategy of ESSs, and the other is the capacity allocation of ESSs. However, the relevant research on the application of ESSs in ports is not yet mature, and even in many aspects is still in the initial stage.

The coordination strategy of ESS is a key issue in improving port energy efficiency, reducing carbon emissions, and promoting clean energy consumption. The main technical means in current research include improving the operation efficiency of port areas and adopting clean energy to replace traditional energy. At present, improving the

operation efficiency of port areas is mainly achieved by optimizing the port traffic scheduling strategy, such as berth scheduling [7], berth and quay crane (B-QC) joint scheduling [8], yard crane scheduling [9], transfer vehicle scheduling [10], water-rail joint scheduling [11]. However, the primary purpose of traditional port traffic scheduling management is to achieve logistics optimization. Most studies did not consider the coupling relationship between logistics and energy consumption, resulting in limited improvement of energy efficiency and carbon emissions in port areas.

Replacing traditional energy with clean energy is another effective way to reduce port carbon emissions [12–15]. The coordinated method of ESSs is the key to clean energy consumption in ports. In [12], to improve the energy efficiency of ports, a two-stage sequential scheduling method for ports was proposed. The first stage was to optimize the berth and yard operation, and on this basis, a port energy management method was proposed. In [13], the coordination problem of multiple

types of loads in ports was addressed by integrating high-power and high-energy density ESSs. A coordinated strategy for port energy consumption was proposed to improve port operational performance. However, Ref. [12,13] did not consider changing port loads through traffic scheduling. That is, they did not consider the “transport-energy” coupling characteristics of ports to address the clean energy consumption. Some studies achieve load coordination through special means. In [14], a hierarchical structure of multi agent systems was proposed to achieve real-time control of port flexible loads to promote the consumption of renewable energy in ports. Some studies focused on hydrogen consumption in ports. In [15], based on port logistic activities, a life cycle assessment and total cost of ownership (TCO) model of hydrogen fuel vehicles in ports was formulated, and the energy, economic and environmental performance of hydrogen utilization under various green hydrogen and gray hydrogen in ports is further discussed. However, it only discussed the hydrogen energy consumption of mobile equipment in ports, and the energy coordination in the process of hydrogen production from electricity in ports was not considered. At present, most studies on clean energy consumption in ports only consider the coordination methods of power load, hydrogen load, cold or heat load, and rarely consider coordinating energy imbalance through the transferability characteristics of multiple energy loads in ports. That is, demand side management of loads can be realized by considering the coupling characteristics of “transportation-energy” in ports. With the update and development of port equipment, port equipment equipped with clean energy will dominate energy consumption in ports. Joint consideration of the port transportation and energy will have a significant optimization potential in improving energy efficiency and reducing carbon emissions. However, there are few studies involving this aspect.

Due to the high capital cost and limited lifetime of ESSs, reasonable capacity allocation of ESSs is another key issue to coordinate the imbalance of energy consumption economically and efficiently in ports. In [5], 17 possible renewable energy and ESSs solutions for seaports were evaluated through energy levelization cost and environment cost to determine the capacity and operation strategy of a near-zero carbon port. However, the enumeration method used in this study only verifies the effectiveness of the hybrid renewable energy system, which is difficult to achieve optimal performance. In [16], to address the optimal allocation and operation of renewable energy and ESS in ports, a two-stage optimization model was established to optimize the investment and operation cost of the system. However, it did not consider the effect of the actual operation strategy on the system capacity in the capacity optimization stage. In [17], to provide energy for the cold ironing system locally, a capacity method for PV plants and ESSs in ports was proposed with the goal of optimizing annualized life cycle cost of ports. However, it only considers the cold ironing, without considering other key electrical and hydrogen loads in ports. In [18], a hybrid energy storage capacity allocation method was proposed to coordinate electric/hydrogen load in ports with the goal of optimizing daily operating cost. However, it only considered the daily operation cost of ports for configuration, and did not consider the life cycle cost of the port. In addition, changes in the port's electricity/hydrogen load were not considered. In the issue of ESS coordinating energy consumption in seaports, the capacity configuration of ESS is related to various aspects of seaports, such as scales of seaport, operation modes, and operational strategies. At present, there is not much research on the allocation of ESS capacity in port renewable energy consumption. Few studies have considered both economic and environmental costs to optimize the configuration of ESSs. Furthermore, due to the mutual influence and constraints between ESS operation strategy and capacity configuration, if the two can be combined to design the capacity configuration of ESS, the results obtained will be more efficient.

To promote the consumption of renewable energy and reduce carbon emissions in ports, in this paper, considering the coupling characteristics of transportation and energy, a multi-energy load model of ports is formulated. Furthermore, due to the mutual influence and constraint

between the operation strategy and capacity configuration of ESSs, a hybrid energy storage system (HESS) energy management and capacity configuration method considering the hydrogen/ electricity load characteristics of ports is proposed. The proposed method optimizes the capacity configuration and operation strategy of HESS by a nested bi-layer model, and improves the solution performance by an improved metamodel - based solution algorithm. The main contributions of this paper are as follows.

- A daily electric load model and a daily hydrogen load model are presented based on the coupling characteristics of “transportation - energy” of ports to realize the demand-side management of multi energy load of ports.

- A day-ahead transportation and energy collaborative scheduling model considering carbon emission costs is proposed to realize the economic and low-carbon operation of ports.

- A HESS capacity allocation model is proposed with the goal of minimizing the annual operational life cycle cost of ports, and an improved metamodel based solution algorithm is presented to improve solution performance.

The rest of this paper is organized as follows. Section 2 introduces an overview of the proposed bi-layer model. Section 3 presents the HESS coordination strategy for port operation. The HESS capacity allocation model is formulated in Section 4. Section 5 discusses the improved metamodel based solving algorithm. Section 6 focuses on the case study and discussion. Finally, the paper is summarized in Section 7.

2. Overview of the proposed method

Container terminals with continuous berths are being studied in this study. Fig. 1 presents an overview of the port area study case. The multi-energy load in this paper mainly considers hydrogen load and electric load. According to the operation process of ports, in the container unloading stage, electric energy consumption of fixed logistics equipment is mainly considered, such as cold ironing and quay cranes. In the stage of transportation to yards, hydrogen energy consumption is considered for mobile logistics equipment such as container trucks and RTGs, as it is more convenient to use hydrogen power for mobile equipment.

In this paper, a HESS is composed of a power to hydrogen (P2H) equipment (electrolyzer (Elz) and hydrogen tank) and a Lithium-ion battery (LiB). The P2H equipment mainly coordinates the hydrogen load. Besides, due to the configuration of hydrogen tank, it also affects the real-time electrical power balance of ports. LiB not only coordinates the balance of renewable energy and electric load in ports but also

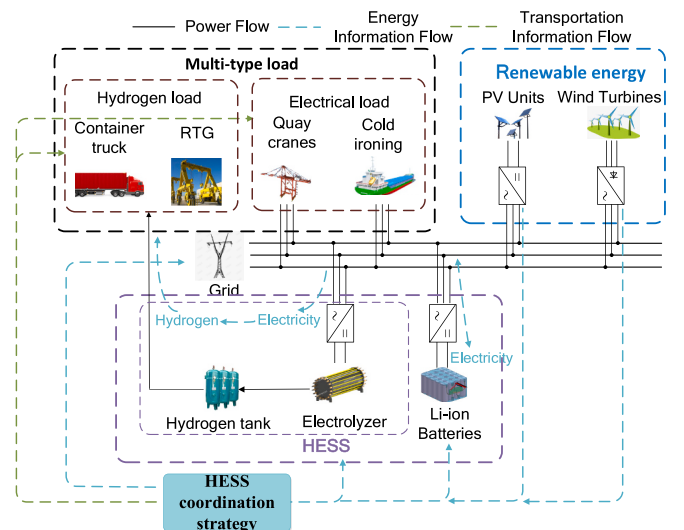


Fig. 1. Overview of the port area study case.

indirectly coordinates the balance of hydrogen energy, because hydrogen energy also comes from electricity. As a result, the hydrogen energy flow and the electricity energy flow in ports affects each other.

Since the hydrogen/electricity load variation of ports is directly related to the port transportation, to coordinate the hydrogen/electricity energy imbalance through HESS more effectively, energy problem optimization can be combined with transportation problems. Fig. 2 shows the relationship between the change of electric load and B-QC scheduling in the container unloading stage. B-QC allocation problems include berth location, berth time, number, and sequence of QC allocation. In Fig. 2, the yellow part shows the change in power load when the berth time is shifted. The blue part shows the change in power load when both the berth time and the number of QCs change. According to Fig. 2, the power imbalance between load and renewables is quite different under different B-QC scheduling results. In the stage of transportation to yards, containers are simultaneously transported to yards when unloaded from vessels, thus the hydrogen consumption of mobile equipment is also affected by B-QC scheduling. As a result, a multi-energy load model based on traffic scheduling can be formulated to realize demand-side management of port loads, to better coordinate the multi energy imbalance in ports.

The daily energy imbalance of hydrogen / electricity in ports can be mitigated through two methods, namely demand-side management of port energy and coordination of HESS. If the two can be considered together, that is, the demand-side management of port energy and the coordination of HESS are considered at the same time, the coordination performance of energy can be further improved, while also enhancing the performance and economy of HESS.

The operation strategy and capacity allocation of HESS are mutually influenced and constrained. Therefore, a nested bi-layer HESS capacity allocation model is proposed, as shown in Fig. 3. By iteratively optimizing the upper layer and lower layer, the bi-layer model optimizes the HESS capacity and operating strategy. In the lower layer, considering the characteristics of hydrogen/electricity load in ports, a “transportation-energy” collaborative scheduling model is formulated. Then, the operating parameters such as the lifetime of HESS and the operating costs of ports are obtained. In the upper layer, a capacity allocation model for HESS is formulated with the goal of minimizing the annual operational life cycle cost of ports. The upper layer and lower layer are iteratively optimized through the HESS capacity parameters and the operating parameters of ports.

In addition, to avoid the problem of low solution accuracy or efficiency caused by the huge computational cost of traditional heuristic solution algorithms, an improved model-pursuing sampling (MPS) algorithm is presented, which improves the accuracy of the solution at a low cost.

3. Lower layer HESS coordination strategy

The lower layer of the proposed method is the HESS coordination

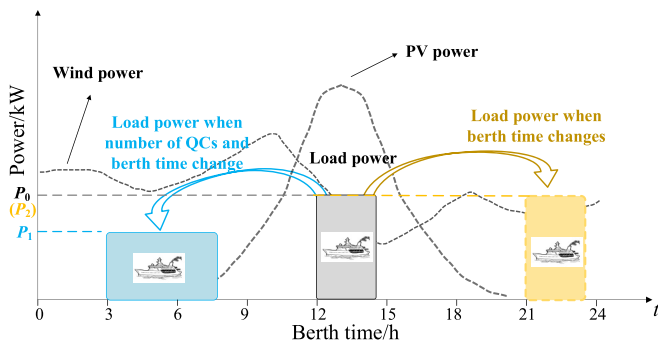


Fig. 2. Schematic diagram between the electric load and B-QC results of a vessel.

strategy. With the electrification / hydrogen power of port equipment, traditional port traffic scheduling also affects the port energy consumption. Therefore, a hydrogen / electricity load model can be formulated combined with the traffic scheduling to realize demand-side management of the hydrogen/electricity load in ports.

A. Multi-energy load model based on port traffic scheduling

The operation process in ports is “unloading - yard transportation”. According to the operation process of ports, the electrical load and hydrogen load are related to the fixed and mobile equipment in ports, respectively.

1) *The electric load model based on ports B-QC scheduling.* The electric load of ports mainly considers the power of quay cranes and cold ironing power, which is mainly related to B-QC scheduling. As a result, an electric load model based on B-QC scheduling can be established.

Because standard container terminals are studied in this paper, the power of each QC can be considered constant in each scheduling cycle. The total power of QCs for vessel i P_{cra} is that

$$P_{cra,i} = p_{cra} \cdot n_{QC,i}, \quad \forall i \in \nu \quad (1)$$

The time in port of vessel i mainly depends on its containers and the number of allocated QCs. The time in port of vessel i t_{ber} is as follows:

$$t_{ber,i} = TEU_i / (n_{QC,i} \cdot v_{cra}), \quad \forall i \in \nu \quad (2)$$

where TEU_i is the number of containers of vessel i .

The departure time $t_{et,i}$ of vessel i is that

$$t_{et,i} = t_{st,i} + t_{ber,i}, \quad \forall i \in \nu \quad (3)$$

During berthing, the electrical load of vessels changes little. For simplicity, the cold ironing power of vessels is set to constant. Therefore, the electric load model of vessel i during berthing can be established as follows:

$$P_{port}(k) = \sum_i P_{iron,i}(k) + \sum_i P_{cra,i}(k) \quad (4)$$

where k is the scheduling cycle.

The B-QC scheduling should meet the following constraints.

1) *Berth position constraints.* This paper focuses on continuous berths. The position of berthing should be within the coastline, namely:

$$x_{end,i} + l_{safe} \leq l_{line}, \quad \forall i \in \nu \quad (5)$$

The end position of the vessel can be obtained as follows:

$$x_{end,i} = x_{sta,i} + l_{ves,i}, \quad \forall i \in \nu \quad (6)$$

2) *Berthing time constraints.* The actual berthing time should be within a certain range of the expected berthing time:

$$T_{wait}^{min} < t_{st,i} - T_{esti,i} < T_{wait}^{max}, \quad \forall i \in \nu \quad (7)$$

3) *No two vessels are allowed to overlap in time and space.* That is, if $(x_{sta,i} - x_{end,j}) \cdot (x_{end,i} - x_{sta,j}) < 0$, $\forall i, j \in \nu$, then

$$\begin{cases} t_{et,i} < t_{st,j}, & \text{if } t_{st,i} \leq t_{st,j} \\ t_{et,j} < t_{st,i}, & \text{if } t_{st,i} > t_{st,j} \end{cases}, \quad \forall i, j \in \nu \quad (8)$$

4) *The number of QCs allocated to vessel i should meet:*

$$n_{QC,i}^{min} < n_{QC,i} < n_{QC,i}^{max}, \quad \forall i \in \nu \quad (9)$$

5) *The total number of QCs allocated in any optimization cycle k should meet:*

$$\sum_i n_{QC,i}(k) \leq n_{QC}^{max}, \quad \forall i \in \nu \quad (10)$$

6) *Any QC is not allowed to cross operation at the same time k :*

$$(x_{QC,m}(k) - x_{QC,n}(k)) \cdot (x_{sta,i} - x_{sta,j}) \cdot S_{m,i}(k) \cdot S_{n,j}(k) \geq 0, \quad \forall m, n \in Q, \quad \forall i, j \in \nu \quad (11)$$

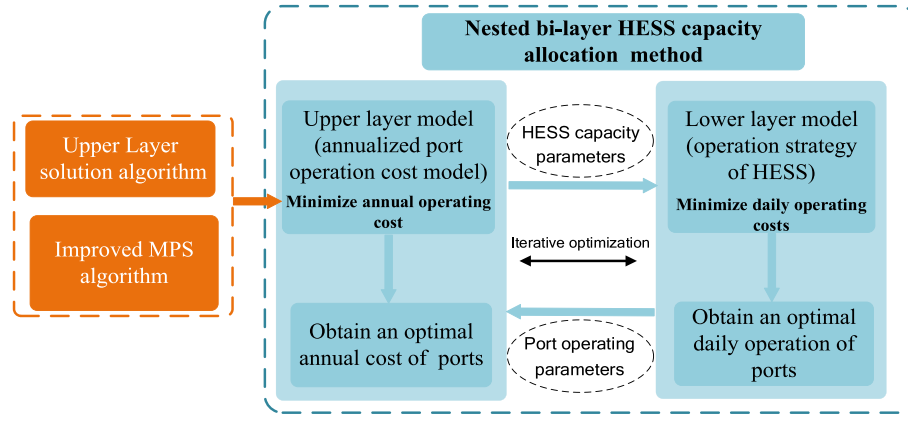


Fig. 3. Bi-layer model frame structure.

7) Vessels should complete berthing and departure within a day since the scheduling period is one day in this paper:

$$t_{et,i} \leq 24, \quad \forall i \in \nu \quad (12)$$

The B-QC scheduling model includes continuous variables (t_{sb} , x_{sta}) and 0–1 variables $S_{m,i}(k)$. Furthermore, the port electric load that requires HESS coordination can be obtained through the B-QC scheduling model.

2) *Hydrogen load model based on port yard scheduling.* According to the operation process of ports, containers are transported to yards simultaneously when unloading from vessels. In yard transportation stage, the energy consumption of containers is affected by both B-QC scheduling and yard scheduling. Therefore, the hydrogen load in ports mainly considers the consumption of container trucks and RTGs in this stage. A hydrogen load model based on yard scheduling can be established based on the container transportation routes in ports. The schematic diagram of a port yard and container route is shown in Fig. A1 in the Appendix.

The required yards for the container of vessel i is as follows:

$$n_{ves,i} = \text{ceil} \left(TEU_i / n_{TEU}^{\max} \right) \quad (13)$$

The ceil (x) function means the smallest integer greater than or equal to x .

1) *Hydrogen consumption model of container trucks.* The hydrogen consumption per unit distance of container trucks is fixed and known. Therefore, according to Fig. A1 in the Appendix, the transportation distance D can be obtained through the berth position of vessel i and the corresponding yard location A_i . The hydrogen consumption per unit time of container trucks that serving vessel i $V_{trk,i}$ is that

$$V_{trk,i} = \sum_j (v_{trk} \cdot D_j \cdot n_{TEU,j}) / t_{ber,i} \quad (14)$$

According to the operation process of ports, the hydrogen consumption time sequence of vessel i during berthing can be obtained from its berthing time sequence and eq. (14).

2) *Hydrogen consumption model of RTGs.* Generally, the energy consumption of RTG is calculated by the energy consumption per unit container. The hydrogen consumption of RTGs that serving vessel i per unit time $V_{RTG,i}$ is that

$$V_{RTG,i} = \sum_j (v_{RTG} \cdot n_{TEU,j}) / t_{ber,i} \quad (15)$$

Similarly, the RTGs hydrogen consumption time sequence of vessel i during berthing can be obtained.

The constraints for yard scheduling are as follows.

1) Each yard can serve up to one vessel, that is, for any vessel i , each yard should satisfy:

$$\sum_i S_{i,j} \leq 1, \quad \forall i \in \nu, j \in y \quad (16)$$

2) The total yards required for vessel i should meet:

$$n_{ves,i} = \sum_j S_{i,j}, \quad \forall i \in \nu, j \in y \quad (17)$$

3) The number of containers of vessel i should be equal to the sum of the containers allocated to the corresponding yards:

$$\sum_{n_{ves,i}} n_{TEU,j} = TEU_i, \quad \forall i \in \nu, j \in y \quad (18)$$

The optimization variables of port yard scheduling include the yard location $S_{i,j}$ of vessel i and the corresponding number of containers $n_{TEU,j}$. Furthermore, the hydrogen load model that requires coordination with P2H equipment can be obtained through the yard scheduling model.

B. *Day-ahead collaborative scheduling strategy for transportation and energy in ports*

Based on the traffic scheduling model formulated in previous section, in this section, considering carbon emission cost in ports, a “transportation-energy” collaborative scheduling model is formulated to improve the operation economy and reduce carbon emissions in ports.

The goal of energy scheduling is to coordinate the supply and demand imbalance of hydrogen/electricity in ports through a HESS. Further, the scheduling strategy improves the operation economy and promotes the consumption of renewables. Therefore, the economic operation model of the day-ahead scheduling model can be formulated as follows (day-ahead scheduling cycle T_d is 1 h):

$$\min C_{ope} = \sum_{k=1}^{24} (C_{elz}(k) + C_{bat}(k) + C_{ves}(k) + C_{grid}(k)) + \sum_{k=1}^{24} (c_{penal} \cdot P_{grid}(k) \cdot T_d) \quad (19)$$

In cost function (19), the operation cost of the port area includes the daily operation cost of HESS, the vessel operation cost, and the main grid cost. To promote the consumption of renewables, a penalty cost for using the main grid is introduced, that is the second item of Eq. (19). But it is not included in the actual operating costs of the port area. The various costs are as follows.

1) *Operating cost of LiB C_{bat} .* The operating cost of LiB mainly includes its daily degradation cost. The degradation of LiB has a nonlinear relationship with its depth of discharge (DOD). In this paper, the leveled cost of energy model is used to describe its degradation cost, which is shown in previous research of authors [19].

2) *Operating cost of Elz C_{elz} .* The operating cost of Elz includes the

degradation cost and the startup/shutdown cost. The lifetime of Elz is generally related to its service time, which is generally 10,000–25,000 h. As a result, the operating degradation cost in each scheduling cycle can be obtained according to the lifetime-time linear model of Elz:

$$C_{elz} = \sum_{i=1}^{24} (C_{cap,elz}/T_{lft,elz} + C_{om}) \cdot \delta_i + C_{stup} \cdot \gamma_{on,i} + C_{shud} \cdot \gamma_{off,i} \quad (20)$$

3) Operating cost of vessels C_{ves} . The daily operating cost of vessel i $C_{ves,i}$ includes the cost of waiting for berthing and the cost of CO₂ emissions due to waiting for berthing:

$$C_{ves,i} = \sum_i C_{penal}^{ws} \cdot t_{wait,i} + \sum_i C_{berth,i}^{co2} \cdot t_{wait,i}, \quad \forall i \in \nu \quad (21)$$

The CO₂ emissions of vessel i in T_d can be obtained by fitting the energy consumption of vessel auxiliary generator:

$$E_{ves,i} = (\alpha_0 + \alpha_1 \cdot P_{iron,i} + \alpha_2 \cdot P_{iron,i}^2) \cdot T_d \quad (22)$$

where $\alpha_0, \alpha_1, \alpha_2$ are the fitting coefficients for the fuel consumption of vessel auxiliary generators [14].

When waiting for berthing, if the carbon emissions of vessel i exceeds its per unit time carbon emission limit, a CO₂ emission fee should be paid:

$$C_{berth,i}^{co2} = c_{co2} \cdot (E_{ves,i} - E_{ves,i}^{\max}) \quad (23)$$

where $E_{ves,i}^{\max}$ is the carbon emission limit of vessel i per unit time.

4) Main grid cost C_{grid} . C_{grid} includes electricity costs and carbon emissions costs (the cost of feedback energy to the main grid is not considered in this paper):

$$C_{grid}(k) = c_{grid} \cdot P_{grid}(k) \cdot T_d + c_{grid}^{co2}(k) \cdot T_d \quad (24)$$

Similarly, if the carbon emission of the main grid exceeds the carbon emission limit of the port area per unit time, the carbon emission cost is that

$$C_{grid}^{co2}(k) = c_{co2} \cdot (E_{grid}(k) - E_{grid}^{\max}) \quad (25)$$

where $E_{grid}^{\max}$ is the carbon emission limits of the main grid per unit time. E_{grid} is the carbon emissions as follows [14]:

$$E_{grid}(k) = \alpha_{co2} \cdot P_{grid}(k) \cdot T_d \quad (26)$$

The constraints that Eq. (19) should satisfy are as follows.

1) The electricity generated and consumed in ports should be balanced:

$$P_{pv}(k) + P_{wt}(k) + P_{bat}(k) + P_{elz}(k) + P_{grid}(k) = P_{port}(k) \quad (27)$$

2) Each part should meet its own power constraints:

$$\begin{cases} P_{bat}^{\min} \leq P_{bat}(k) \leq P_{bat}^{\max} \\ P_{elz}^{\min} \leq P_{elz}(k) \leq P_{elz}^{\max} \\ P_{grid}^{\min} \leq P_{grid}(k) \leq P_{grid}^{\max} \end{cases} \quad (28)$$

3) Energy constraints of the HESS:

$$\begin{cases} SOC_{bat}^{\min} \leq SOC_{bat}(k) \leq SOC_{bat}^{\max} \\ LOH_{tank}^{\min} \leq LOH_{tank}(k) \leq LOH_{tank}^{\max} \end{cases} \quad (29)$$

where SOC_{bat} is the state of charge (SOC) of LiB. LOH_{tank} is the level of hydrogen (LOH) of the hydrogen tank, can be obtained as follows:

$$LOH_{tank}(k+1) = LOH_{tank}(k) - (P_{elz}(k) \cdot T_d \cdot \eta_{elz}) / W_{H2} \quad (30)$$

4) Due to different daily operating conditions, it is necessary to constrain the energy of HESS at the end of each day to a reasonable range:

$$\begin{cases} \kappa_{st}^{bat} \cdot SOC_{bat}(0) \leq SOC_{bat}(24) \leq \kappa_{et}^{bat} \cdot SOC_{bat}(0) \\ \kappa_{st}^{tank} \cdot LOH_{tank}(0) \leq LOH_{tank}(24) \leq \kappa_{et}^{tank} \cdot LOH_{tank}(0) \end{cases} \quad (31)$$

Cost function (19) is composed of a traffic scheduling submodel (Eqs. (1) to (18)) and an energy scheduling submodel (Eqs. (19) to (31)), including the traffic scheduling variables ($t_{st}, x_{sta}, S_{m,i}(k), S_{i,j}, n_{TEU,i}$) and the energy scheduling variables ($P_{bat}, P_{elz}, P_{grid}$) of ports. Collaborative scheduling is achieved by coordinating the power load P_{port} and hydrogen load through B-QC scheduling and yard scheduling, that is Eqs. (27) and (30), to obtain better economic operation and carbon emission performance.

By solving (19), the daily operating parameters of the ports can be obtained, which are the lifetime of HESS and the various operating costs of ports. When obtaining the daily power of LiB, the daily SOC variation curve can be obtained. The daily degradation of LiB can be obtained from the daily SOC variation curve according to the rain-flow counting method [20].

Furthermore, the lifetime of LiB can be obtained according to the daily degradation. Specifically, when the daily LiB degradation is obtained, the annualized lifetime degradation of LiB can be calculated and converted into lifetime based on the annual service coefficient:

$$T_{lt,LiB} = T_{lft,LiB} / (T_{day,LiB} \cdot 365 \cdot \phi) \quad (32)$$

The lifetime of Elz $T_{lt,elz}$ is a linear function of time. Similarly, the lifetime of Elz can be calculated based on the annual service coefficient:

$$T_{lt,elz} = T_{lft,elz} / (T_{day,elz} \cdot 365 \cdot \phi) \quad (33)$$

The lower layer operational parameters include ($C_{ope}, C_{ves}, C_{grid}, T_{lt,LiB}, T_{lt,elz}$).

4. HESS capacity allocation model of upper layer

The upper layer, which is the capacity allocation model of HESS, aims to optimize the annual operational life cycle cost of ports when configuring HESS. The annual operating cost model of the port area formulated in upper layer is that:

$$\min C_{port} = C_{hess} + C_{run} \quad (34)$$

1) C_{hess} includes the capital cost C_{cap} , replacement cost C_{rep} , maintenance cost C_{maint} and remanent value C_{rec} :

$$C_{hess} = C_{cap} + C_{rep} + C_{maint} - C_{rec} \quad (35)$$

The capital cost of HESS C_{cap} is that.

$$C_{cap} = (c_{pb} \cdot P_{bat} + c_{eb} \cdot E_{bat} + c_{pe} \cdot P_{elz} + c_{ee} \cdot W_{H2}) \cdot \gamma_{CRF} \quad (36)$$

$$\gamma_{CRF} = (1 + \gamma)^{T_n} / ((1 + \gamma)^{T_n} - 1) \quad (37)$$

The HESS replacement cost in T_n is that

$$C_{rep} = [(c_{pb} \cdot P_{bat} + c_{eb} \cdot E_{bat}) \cdot n_{bat} + (c_{pe} \cdot P_{elz}) \cdot n_{elz}] \cdot \gamma_{CRF} \quad (38)$$

The replacement times of LiB and Elz in T_n can be obtained from their lifetime:

$$n_{x=bat,elz} = \text{ceil} \left(T_n / \left(T_{lt,x} - 1 \right) \right) \quad (39)$$

The remanent values of HESS is that

$$\begin{aligned} C_{rec} = & [(c_{pb,r} \cdot P_{bat} + c_{eb,r} \cdot E_{bat}) \cdot (n_{bat} + 1) + \\ & (c_{pe,r} \cdot P_{elz}) \cdot (n_{elz} + 1)] \cdot \gamma_{CRF} \end{aligned} \quad (40)$$

The maintenance cost of HESS is that

$$C_{\text{maint}} = c_{\text{bat}}^{\text{maint}} \cdot E_{\text{bat}} + c_{\text{elz}}^{\text{maint}} \cdot P_{\text{elz}} + c_{\text{tank}}^{\text{maint}} \cdot W_{H^2} \quad (41)$$

2) The annual operating cost of the port C_{run} . C_{run} includes annual berth costs and main grid costs:

$$C_{\text{run}} = (C_{\text{ves}} + C_{\text{grid}}) \cdot 365 \cdot \phi \quad (42)$$

In HESS capacity allocation model, the capacity of LiB and Elz must meet certain constraints to meet the power and energy balance requirements for the daily operation of the lower layer.

1) Constraints of minimum power capacity of Elz. The hydrogen generated by the Elz operating at minimum power per day should meet the hydrogen load demand for at least one day:

$$\sum_{k=1}^{24} P_{\text{elz}} \cdot T_d \cdot \eta_{\text{elz}} \geq \sum_i (V_{\text{trk},i} + V_{\text{RTG},i}) \quad (43)$$

2) To meet the one-day electricity balance in day-ahead scheduling, the power capacity of HESS should meet the maximum power deviation:

$$P_{\text{elz}} + P_{\text{bat}} \geq \max(P_{\text{pv}} + P_{\text{wt}} + P_{\text{grid}}^{\text{min}}) \quad (44)$$

The upper layer optimization variables include the capacity allocation parameters ($P_{\text{bat}}, E_{\text{bat}}, P_{\text{elz}}, W_{H^2}$) of HESS. The iterative optimization of the bi-layer model is realized by exchanging the capacity parameters in upper layer model and the operating parameters in lower layer model.

5. Improved solution method of upper layer model

In HESS bi-layer model, there are large number of variables in the lower layer and the calculation is intensive. The upper layer model is a multivariable nonlinear model. As a result, when solving the upper layer model, the traditional heuristic algorithm (such as particle swarm optimization, PSO) needs to call a large number of iterative calculations of the lower layer. There is a huge computational consumption problem, which is easy to lead to low accuracy or low efficiency. The mode-pursuing sampling (MPS) algorithm [21] based on meta-model can reduce the call of the lower layer by constructing a meta-model, which is an effective method to solve the computationally intensive black-box problem.

The key of the meta model-based algorithm is to construct a mathematical model similar to the source model through a special sampling method, so as to replace the original complex model. It improves solving efficiency by reducing the number of expensive calculations used for evaluation. The schematic diagram of the meta model-based method is shown in Fig. 4. The metamodel based optimization algorithms is particularly suitable for computationally intensive black-box optimization problems and is an effective way to improve design efficiency [21,22].

The metamodel based MPS algorithm was proposed by G.Gary Wang in 2004. It uses mode-pursuing sampling method to generate more sample points in areas where the source model is more likely to have a minimum point. Then, radial basis function (RBF) is used to determine the region of the global optimal solution, and a quadratic polynomial response surface model (RSM) is constructed to determine global convergence.

In MPS algorithm, the convergence speed control factor is obtained through the multi-correlation coefficient (R^2), and the range of the search area is dynamically adjusted through R^2 . However, MPS also has certain limitations, that is, it is easy to converge to a local optimal solution before generating sufficient exploration points [21]. Therefore, the convergence speed control factor is improved to compromise global convergence and convergence speed, thereby improving the overall performance of the solving algorithm.

The correction of the convergence speed control factor includes two steps, one is the correction of the R^2 , the other is the correction of the convergence factor based on the first step.

1) Correction of R^2 . In MPS algorithm, R^2 ($R^2 \in (0, 1)$) is used to evaluate the accuracy of the constructed response surface model. The more R^2 tends to 1, the higher the fitting accuracy of the constructed RSM model is. It is effective when the number of calls to the source model is low. However, as the number of calls increases, the number of variables also increases, which also leads to an increase in R^2 . But it does not necessarily mean a better RSM fitting model. In this case, MPS may converge prematurely before generating enough exploration points and fall into local optimal solution. Therefore, to reduce the impact of the number of variables on the evaluation index, R^2 can be modified as follows:

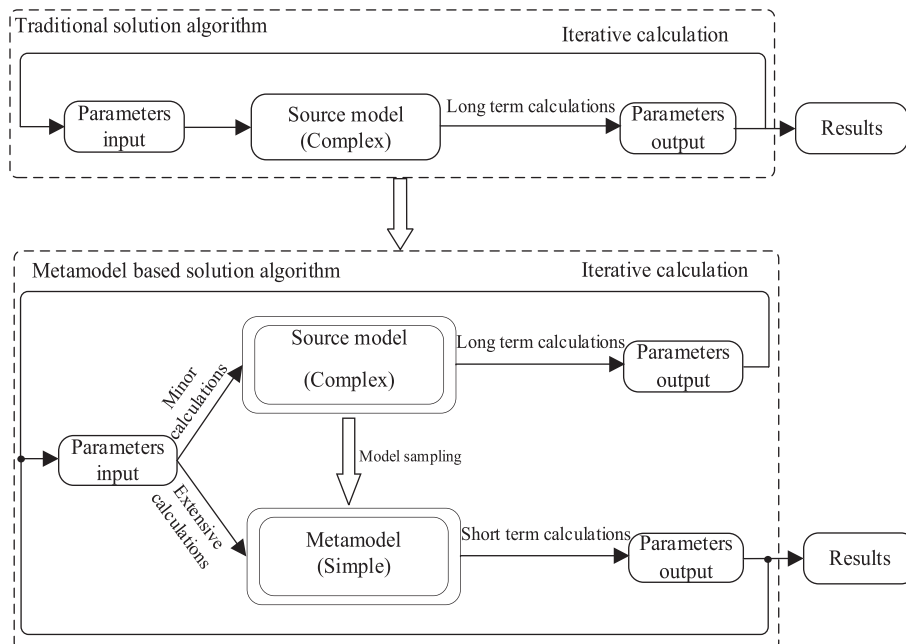


Fig. 4. Schematic diagram of meta model optimization algorithm.

$$R_{adj}^2 = 1 - \left(\frac{\sum_{i=1}^{n_s} (\hat{y}_i - \bar{y})^2}{(n_s - n_t - 1)} \right) / \left(\frac{\sum_{i=1}^{n_s} (y_i - \bar{y})^2}{n_s} \right) \quad (45)$$

where R_{adj}^2 is the modified evaluation index, n_s is the sample size, and n_t is the number of variables. y_i , $\hat{y}_i$, $\bar{y}$ are the true value, estimated value and average value of the model, respectively.

In (45), R_{adj}^2 is dynamically adjusted by introducing the number of variables, reducing the impact of the number of variables on R^2 .

2) Correction of convergence speed control factor. In MPS algorithm, when R^2 increases, the speed control factor r increases to generate more sample points in the current region and accelerate the convergence process. On the contrary, if R^2 decreases, r tends to 1 to expand the search region. Based on it, the Sigmoid function has strong nonlinearity, and its characteristics are consistent with the dynamic changes of r . r can be corrected based on the Sigmoid function as follows:

$$r_{adj} = 2 \cdot r_{max} / \left(1 + e^{-\alpha(R_{adj}^2 - 1)} \right) + 1 \quad (46)$$

where r_{max} is the maximum value of r_{adj} . The experience value is provided for r_{max} in MPS algorithm [21]. α is an adjustment coefficient.

The characteristic curve of Eq. (46) is shown in Fig. 5.

Through the above two steps of correction, the convergence speed control factor is dynamically adjusted to solve the problem that MPS algorithm is easy to enter a local convergence or slow convergence after entering the local region.

The flow diagram of the proposed bi-layer capacity allocation method is shown in Fig. 6. The bi-layer model is iteratively optimized through the HESS capacity parameters and the operating parameters of ports. When the lifetime of HESS in the lower layer is obtained, the annual operating cost model of the upper layer is used to optimize the capacity of HESS. Similarly, when the HESS capacity results of the upper layer are obtained, the optimal operation strategy of the port area is obtained according to the collaborative scheduling model in the lower layer. The results include the optimal HESS capacity and the optimal port operation strategy.

6. Case study

To verify the effectiveness of the proposed coordination strategy and capacity allocation method of HESS, historical data from Ningbo Port, China was used for simulation. The proposed strategy and models are implemented through MATLAB optimization toolbox and CPLEX toolbox. The lower layer collaborative scheduling model can be solved by the CPLEX optimization toolbox. The simulation parameter settings [23,24] are shown in Table I.

A. Results of collaborative scheduling of “transportation-energy” in lower layer

To verify the effectiveness of the multi energy supply and demand

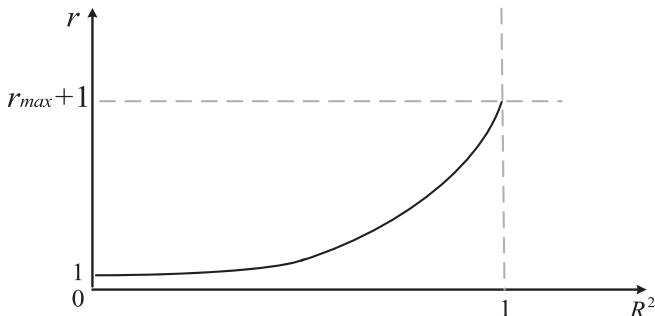


Fig. 5. Characteristic of speed control factor r_{adj} ($\alpha = 5$).

imbalance coordination strategy in the lower layer, a fixed capacity (empirical capacity) of LiB and Elz were used for simulation. The method for determining HESS empirical capacity is shown in previous research of authors [19]. The berth demand information for a typical day in the port area are shown in Table A2 in the Appendix. At the same time, the “transportation-energy” conventional scheduling (CS) method was simulated for comparison [12]. CS method, that is, according to the traditional operation process of ports to complete the transportation scheduling and energy scheduling. First, the B-QC scheduling is implemented with the minimum total berth time, and then the yard scheduling is carried out with the goal of shortest transportation distance. Finally, the HESS coordination method proposed in this paper is used to achieve the energy scheduling. Fig. A3 in the Appendix shows the traffic scheduling results under the proposed collaborative scheduling (simultaneous scheduling, SS) method and CS method. Fig. 7 shows the day-ahead energy scheduling results under the two methods. Table II shows the comparison results of various methods.

According to Fig. A3, Fig. 7 and Table II, both methods can meet the basic requirements of port traffic scheduling, such as intraday berths, no conflict between time and space, and reasonable distribution of QCs. However, the port operation has a better performance in multiple aspects under the SS method. In terms of energy consumption in ports, considering the coupling characteristics of “transportation” and “energy”, the traffic scheduling results under the SS method integrate the energy consumption caused by berth location, QC quantity, and yard location. Compared with CS method, the B-QC scheduling results under SS method are more concentrated in terms of berth location and berth time. It reduces the electrical / hydrogen load fluctuations in ports, while reducing its peak energy consumption, thereby reducing the power fluctuations that HESS needs to coordinate. Although the SS method increases the total berthing time, the total daily power demand in the ports is reduced by 11.9 % compare to CS method. In addition, the total distance transported by the container truck is reduced by 18.6 %, thus reducing the total hydrogen consumption in ports in a day. The change in hydrogen/electricity load in the port area also verifies the goal of multi-energy load-side management through “transportation-energy” collaborative scheduling.

In terms of HESS coordinating the supply and demand imbalance of hydrogen/electricity in ports, as shown in Fig. 7d, the change in SOC under SS method is greater than that under CS method. It indicates that the imbalance between energy supply and demand under SS method is more reliant on LiB for coordination, which also leads to a 17.1 % increase in the daily cost of LiB. Although the berth cost (increased by USD 153) and LiB cost (increased by USD 487) are increased under SS method, the dependence on the main grid is reduced. As a result, the main grid cost under SS method is only 28.6 % (reduced by USD 1053) of that under CS method, thus reducing the total operation cost by 7.2 % compared with CS method. In the lower layer, the transportation and energy are combined to optimize the operation of the port area. It not only reduces the operation cost but also reduces the power disturbance to the main grid, which can effectively promote the consumption of renewable energy in ports.

Further, in terms of carbon emissions in ports, the day-ahead scheduling considers both the operation costs and the environmental costs. The cost function (19) includes the carbon emission costs of port transportation scheduling and energy scheduling. Under the SS method, the energy autonomy performance of the port area is better, which significantly reduces the carbon emissions of the main grid. Although it increases carbon emissions during berthing, the total carbon emissions of the port area are 35 % lower than those of CS method. In addition, to balance the multi-energy imbalance in ports, both SS method and CS method have significantly improved the economic and carbon emission performance compared with using only the main grid.

B. Results of HESS capacity allocation of the upper layer

The proposed HESS bi-layer capacity allocation method was used for capacity allocation with two operation strategies. The parameter

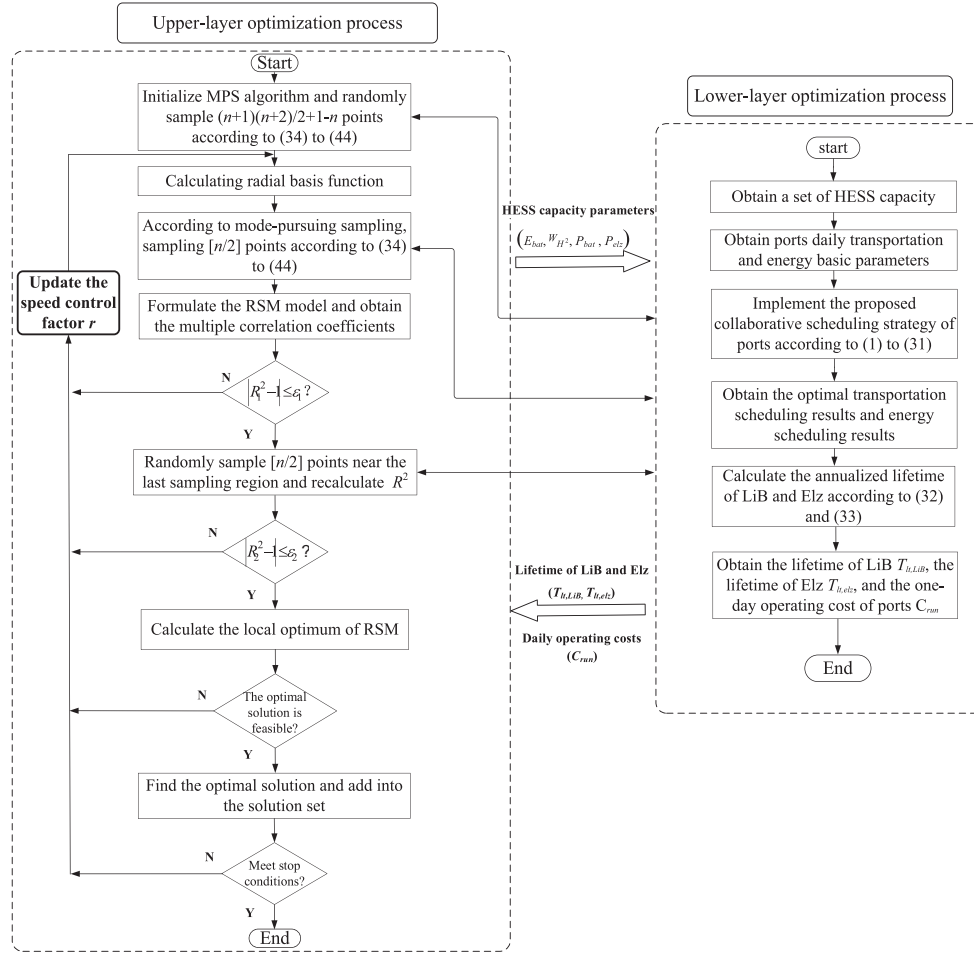


Fig. 6. The flow diagram of the proposed method.

Table I

Parameter settings.

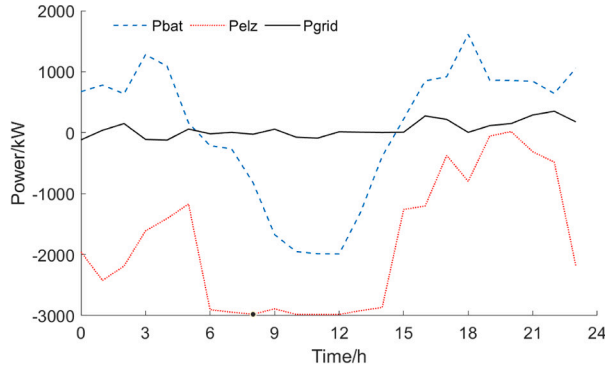
Variables	Values	Variables	Values
a. Day-ahead traffic scheduling parameters			
n_{QC}^{max}	7	l_{safe}	10
L_{line}	1000	n_{TEU}^{max}	288
$T_{wait}^{min}, T_{wait}^{max}$	-3, 6	v_{irk}	1.68
v_{cra}	30	v_{RTG}	4.8
p_{cra}	320		
b. Day-ahead energy scheduling (fixed HESS capacity)			
P_{bat}, E_{bat}	2000, 20,000	c_{co2}	0.15
P_{elz}	3000	$SOC_{bat}^{max}, SOC_{bat}^{min}$	0.8, 0.2
W_{H^2}	5000	$LOH_{tank}^{min}, LOH_{tank}^{max}$	0.8, 0.2
p_{grid}^{max}	5000	η_{elz}	0.23
$C_{cap, bat}$	4,360,000	φ	0.7
$C_{cap, elz}$	1,250,000	$T_{lft, elz}$	21,000
c. HESS capacity configuration parameters			
c_{pb}, c_{eb}	180, 200	$c_{bat}^{maint}, c_{elz}^{maint}$	7, 7
c_{pe}, c_{ee}	250, 100	c_{tank}^{maint}	7
$c_{pb, r}, c_{eb, r}$	18, 20	T_{RT}	20
$c_{pe, r}$	10	γ	0.05
d. Improved MPS algorithm			
λ	50	α	5

settings are shown in Table I. c and Table I. d. The results are shown in Table III, and the optimal operating results under the optimal HESS capacity are shown in Fig. 8.

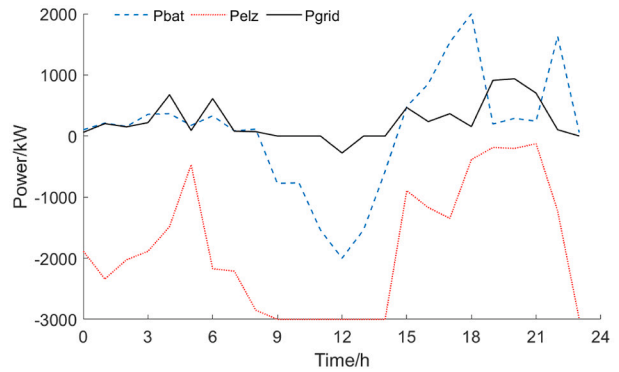
According to Table III and Fig. 8, the optimal capacity of P2H equipment (Elz and hydrogen tank) under the two scheduling strategies

is basically similar. This is because the P2H equipment only needs to coordinate the hydrogen supply and demand balance, and the total hydrogen consumption in a day is similar. Both energy capacity and power capacity of LiB under SS method are higher than that of CS method. This is because the SS method optimizes the economy of operation based on the coupled characteristics of transportation and energy. It relies more on LiB to coordinate the hydrogen/ electricity imbalance in ports, resulting in the increase of LiB capacity allocation. It brings two advantages, reducing the use of the main grid and the total carbon emissions. The daily carbon emissions under SS method is 16.3 % lower than that of CS method. The increase in LiB capacity increases the capital cost. However, it needs to be replaced three times throughout the project period while under CS method, it needs to be replaced four times. It resulted in a 7.9 % reduction in the total annual cost of the port area compared to the CS method.

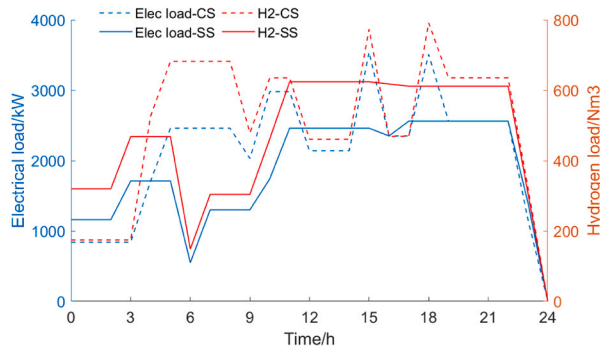
According to Fig. 8, the coordinated results of HESS under the optimal capacity are similar to those in Section 6. A. Due to the reduction of LiB energy capacity, the energy feedback to the main grid increases during the peak period of renewable energy generation, resulting in the power interaction between the port area and the main grid higher than that in Section 6. A. Under the optimal HESS capacity, the daily operating cost under SS method is slightly lower than that in Section 6. A. It is mainly due to the cost reduction caused by the reduction of LiB capacity, but it also slightly increases the total carbon emissions of the port area. In practice, the operating parameters such as the penalty coefficient of the main grid can be reasonably compromised according to the expected carbon emissions, energy self-consistency rate, and other operating indexes of the port area.



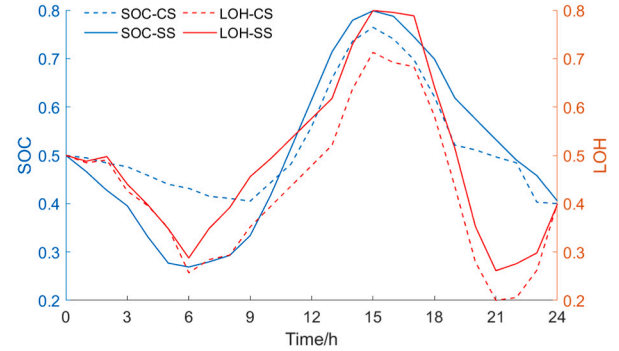
(a) The power of each part under SS method



(b) The power of each part under CS method



(c) Comparison of hydrogen/electricity loads of the two methods



(d) Comparison of SOC and LOH of the two methods

Fig. 7. Comparison of day-ahead scheduling results of the two methods.

Table II

Comparison results of various methods.

		SS method	CS method	Grid only*
Berth time	Total waiting time for berth/h	7.17	0	0
	Total berth time/h	51.25	33.14	33.14
Total transportation	distance of container trucks/km	824.4	1013	1013
Daily operation costs	LiB costs/\$	3321	2834	—
	Elz costs/\$	1428	1428	1428
	Main grid costs/\$	422	1475	27,801
	Berth costs/\$	153	0	0
	Total costs/\$	5324	5737	29,229
Carbon emissions	Carbon emissions of vessels/kg	1562	0	0
	Carbon emissions of main grid/kg	1237	4308	68,670
	Total carbon emissions/kg	2799	4308	68,670

* Renewables and LiB is not configured. The day-ahead scheduling adopts the main grid and P2H, and is realized based on traffic scheduling of CS method.

C. HESS capacity results in different cases

Furthermore, to verify the bi-layer capacity allocation method and analyze the impact of different types of ESSs on energy coordination in ports, different types of ESSs were used to coordinate the energy imbalance. The various configuration cases of ESS are as follows: 1) Case 1, using the proposed HESS capacity allocation and operation strategy; 2) Case 2, only LiB is configured to coordinate power imbalance and P2H equipment is not configured. Hydrogen fuel can only be purchased to meet port demand; 3) Case 3, only P2H equipment is configured, and LiB is not configured. The supply and demand of hydrogen and electricity in ports are respectively completed through P2H equipment and the main grid. The results of HESS capacity allocation for three cases are shown in

Table III

Comparison of HESS capacity allocation results of the two methods.

		SS method	CS method
HESS capacity	P_{bat}/kW	1850	1515
	E_{bat}/kWh	17,960	14,507
	P_{elz}/kW	2566	2436
	W_{H_2}/Nm^3	4452	4505
Operating costs	Daily operation costs/\$	5164	5418
	Annual operating costs / million USD	1.62	1.76
Daily carbon emissions / kg		2907	3477
LiB lifetime / year		5.16	4.53

Table IV. Fig. 9 shows the energy scheduling results under the optimal ESS capacity in three cases.

According to Table IV and Fig. 9, there are significant differences in capacity allocation, operation economy and carbon emissions under three cases. In terms of ESS capacity allocation, the capacity of LiB and P2H devices in Case 1 is significantly larger than that of single ESS case (Case 2 and Case 3). In Case 1, the hydrogen/electricity imbalance in ports is coordinated through HESS. While LiB only needs to coordinate the electricity imbalance in Case 2, and P2H devices only need to coordinate the hydrogen imbalance in Case 3. The coordination of multiple types of loads not only increases the complexity of HESS coordination but also increases the capacity of HESS.

While in terms of the operation economy of the port area, the daily operation cost and annual operation cost in Case 1 are significantly lower than those of the other two cases. Because only the electricity imbalance needs to be coordinated, the energy consumption of the main grid in Case 2 is very small, and most of the renewable energy is fed back to the main grid. The total cost of LiB and the main grid is USD 3810 in Case 1, which is USD 2213 higher than in Case 2. However, the cost of

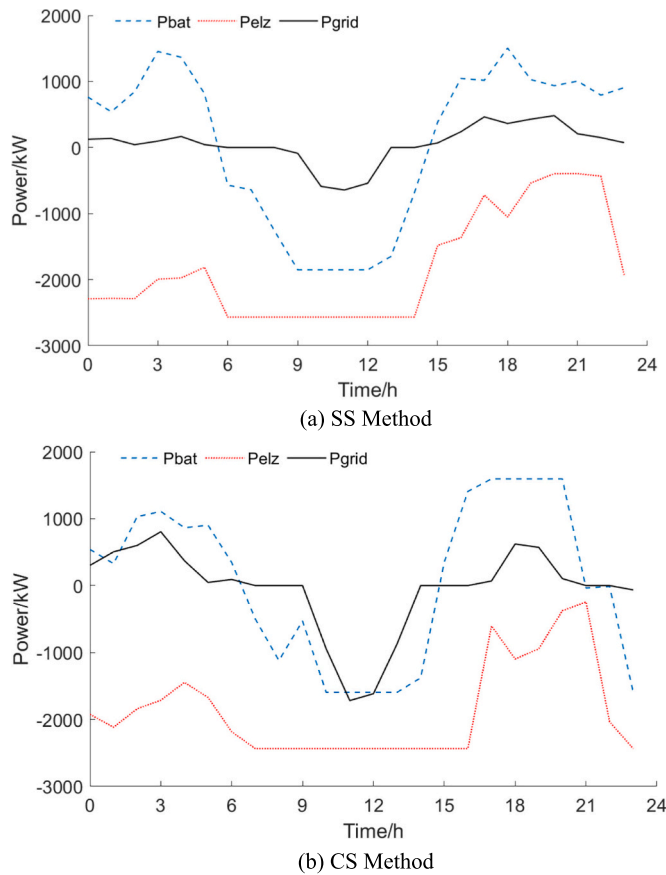


Fig. 8. Optimal operation results of HESS under optimal capacity.

Table IV
Comparison of HESS capacity allocation results under three cases.

		Case 1	Case 2	Case 3
HESS optimal capacity	$P_{bat}/(\text{kW})$	1850	1132	—
	$E_{bat}/(\text{kWh})$	17,960	5142	—
	$P_{elz}/(\text{kW})$	2566	—	2333
	W_{HP}/Nm^3	4452	1763	2241
HESS lifetime	LiB lifetime /Year	5.16	5.15	—
	Elz lifetime /Year	3.42	—	3.42
Port operation costs	Daily operation costs/\$	5164	10,194	26,493
	Annual operation costs/million USD	1.62	2.54	6.12
Total carbon emissions/kg		2907	2370	68,670

hydrogen fuel in Case 2 is USD 8447, resulting in a sharp increase in its daily operating costs.

In terms of carbon emissions, due to that LiB only needs to coordinate the electrical load in Case 2, the dependence on the main grid is lower, reducing the total carbon emissions of the port area. Case 3 is the least economical case in three cases. The main grid is fully used for hydrogen/electricity energy supply, and its economic cost and carbon emissions have obvious disadvantages compared to the configuration of ESSs. The results also verify the effectiveness of the proposed HESS capacity allocation method.

D. Comparison results of different solving algorithms

To verify the effectiveness of the improved MPS algorithm, several commonly used optimization algorithms are selected to solve the proposed upper layer model, such as evolutionary algorithm (genetic algorithm, GA), swarm intelligence algorithm (particle swarm optimization algorithm, PSO), and simulated annealing (SA) algorithm. The initialization parameters of various algorithms are set as follows:

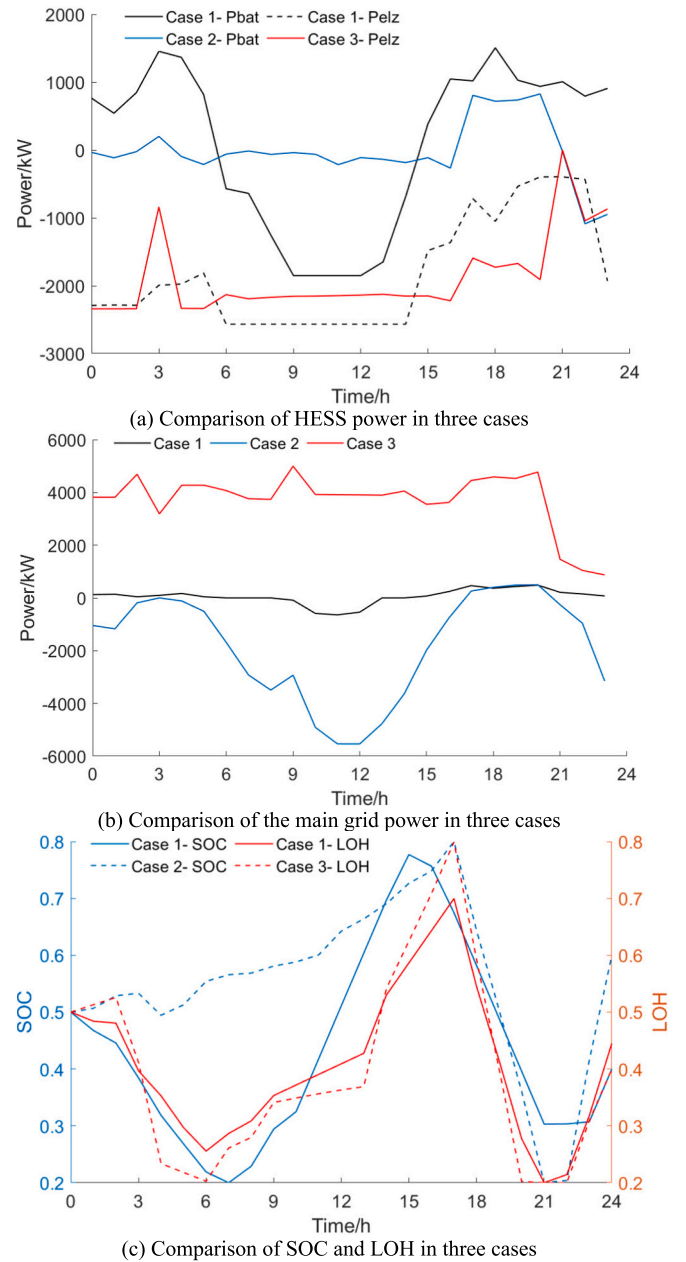


Fig. 9. Operation results under the optimal ESS capacity in three cases.

GA: the population size is 20, and the cross-mutation rate is 0.01; PSO: the particle swarm size is 20, the learning factor is 1.5, and the maximum and minimum inertia weights are 0.8 and 0.4 respectively; SA: the initial temperature is 100, the annealing coefficient is 0.98, and the step size factor is 0.01. The number of iterations of each algorithm is

Table V
Calculation results of each solving algorithm.

Solving algorithm	Annual operating costs/million USD	Average calculation time/min	Average calls of lower layer model
Improved MPS	1.62	58	152
Standard MPS	1.65	53	138
GA	1.70	362	1005
PSO	1.69	362	1011
SA	1.76	361	1004

set to 50. The various algorithms are calculated five times to get the best results. Table V presents the optimal results of different algorithms.

It can be seen from Table V that the annual operating cost of the port area under the improved MPS algorithm is lower than that of other algorithms. Compared with the standard MPS algorithm, GA algorithm, PSO algorithm and SA algorithm, the annual operating cost is reduced by USD 31,044, USD 81,548, USD 77,463, and USD 145,268, respectively. Besides, the average number of calls of the lower layer is only 1/6 of the traditional GA algorithm, PSO algorithm, SA algorithm, which is close to that of the standard MPS algorithm. The comprehensive performance of the improved MPS algorithm is better. In this paper, the convergence speed control factor is improved through a two-step correction. First, by modifying the multi-correlation coefficient (R^2) to reduce the impact of variables increase on the accuracy evaluation of RSM fitting models. Then, based on the modified R^2 , the convergence speed is adjusted through the nonlinear Sigmoid function to compromise the balance between the global search and the convergence speed. As a result, compared with the standard MPS algorithm, the improved MPS algorithm increased 14 calls of the lower layer model. When r_{adj} is low, the search region is expanded to evaluate the global optimal region and avoid too fast convergence to a local optimal region. When r_{adj} approaches the decision threshold, its convergence is promoted by the nonlinearity of Sigmoid function.

Compared with the standard MPS algorithm, the improved MPS solution speed is basically equivalent. However, if the other algorithms improve the accuracy of the solution, it is necessary to increase the iteration times to increase the evaluation times of the lower layer model. But it will also reduce the efficiency of the solution.

7. Conclusions

To promote the renewable consumption and economic operation of ports, a HESS energy management and capacity allocation method is proposed based on the “transportation-energy” coupling characteristics of ports to coordinate the imbalance of hydrogen/electricity supply and demand in ports. The proposed method optimizes the energy management strategy and capacity allocation of HESS through a nested bi-layer model. In the lower layer HESS energy management, a multi-energy load model based on port traffic scheduling is formulated to realize the demand-side management of port multi-energy load. Furthermore, considering the economic operation costs and environmental costs of ports, a “transportation-energy” collaborative scheduling model for HESS coordination is proposed. In the upper layer, a capacity allocation model for HESS is formulated with the goal of minimizing the annual operational life cycle cost of ports. To consider the mutual impact of HESS operation strategy and capacity configuration, the upper layer and the lower layer iteratively optimize each other through capacity parameters and operational parameters. Finally, to improve the accuracy and efficiency of the model solution, an improved MPS algorithm is presented. The simulation results based on historical data show that compared with the traditional port scheduling method, the proposed collaborative scheduling method with fixed HESS capacity reduces the daily operating cost and carbon emissions of ports by 7.2 % and 35 %, respectively. The annual cost and daily carbon emission of the proposed scheduling method are reduced by 8.2 % and 16.3 %, respectively under the optimal HESS capacity compared with the empirical capacity. At the same time, the annual cost of the improved MPS algorithm is reduced by USD 31,044 and USD 77,463 respectively compared with the standard MPS algorithm and PSO algorithm.

CRedit authorship contribution statement

Jingyuan Yin: Conceptualization. Qunhai Huo: Writing – review & editing. Junqiang He: Conceptualization. Xianghua Peng: Software. Tongzhen Wei: Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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